www.Padasalai.Net www.TrbTnpsc.com

VICTORY TUITION CENTRE, CBE-25

QUARTERLY EXAMINATION 2025 MODEL PAPER 2

Class: XII **MATHEMATICS** **Marks:70**

I. ANSWER ANY 7 (Q.NO10 IS COMPULSORY) 7X2=14

1. solve by Cramer's rule 5x-2y+16=0,3y+x-7=0
2. show that |3z-5+i|=4 represent a circle and find its centre and radius
3. Find the modulus and principal value argument of $\sqrt{3}$-i
4. Find the solution of $2\cos^2x-9\cos x+4=0$
5. For what value of x does $\sin x=\sin^{-1}x$?
6. State the reason for $\cos^{-1}\cos(-\frac{\pi}{6}) \neq -\frac{\pi}{6}$
7. obtain the equation of the circle for which (3,4) and (-2,7) are the ends of a diameter.
8. Find the angle between the line $\vec{r}=(2\hat{\imath}-\hat{\jmath}+\hat{k}+t(\hat{\imath}+2\hat{\jmath}-2\hat{k})$ and the plane and $\vec{r}.(6\hat{\imath}+3\hat{\jmath}+2\hat{k})=8$
9. If $\hat{a},\hat{b},\hat{c}$ are three unit vectors such that $\hat{a}\times(\hat{b}\times\hat{c})=\frac{1}{2}\hat{b}$, find the angle between $\hat{a}$ and $\hat{c}$
10. Find the equation of the tangent to the parabola $y2=6x$ parallel to $3x-2y+5=0$

II. ANSWER ANY 7 (Q.NO20 IS COMPULSORY) 7X3=21

11. if $\text{adj}A=\begin{bmatrix}2&-4&2\\-3&12&-7\\-2&0&6\end{bmatrix}$ find A
12. Solve by matrix inversion method $2x+5y=-2$, $x+2y=-3$
13. If $\arg(z-1)=\frac{\pi}{6}$, $\arg(z+1)=\frac{2\pi}{3}$ prove that $|z|=1$
14. Find the sum of squares of roots of the equation $2x^4-8x^3+6x^2-3=0$
15. find the domain of $\sin^{-1}x+\cos^{-1}x$
16. Find the value of $\tan(\sin^{-1}\frac{3}{5}+\cot^{-1}\frac{3}{2})$
17. Find the equation of hyperbola whose centre(2,5), distance between directrices 15, distance between foci is 20 and transverse axis on y axis
18. Find the equation of tangents to the ellipse $5x2+20y2=100$ which are perpendicular to $x+y+2=0$
19. Find the point where the straight line passes through (-5,7,-4) and (13,-5,2) cuts the xy plane
20. show that $\hat{a}\times(\hat{b}\times\hat{c})+\hat{b}\times(\hat{c}\times\hat{a})+\hat{c}\times(\hat{a}\times\hat{b})=0$.

III. ANSWER THE FOLLOWING QUESTIONS : 7X5=35

21 a) If the system of equations $px+by+cz=0$, $ax+qy+cz=0$, $ax+by+rz=0$ has a non trivial solution and $p\neq a, q\neq b, r\neq c$, prove that $\frac{p}{p-a}=\frac{q}{q-b}=\frac{r}{r-c}=2$ (OR)

b)If $\cos\alpha+\cos\beta+\cos\gamma=\sin\alpha+\sin\beta+\sin\gamma=0$ show that $(i)\cos3\alpha+\cos\beta+\cos3\gamma=\cos(\alpha+\beta+\gamma)$ (ii) $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=\frac{3}{2}$

22 a) prove that $(1+\sqrt{3}i)^n+(1-\sqrt{3}i)^n=2^{n+1}\cos\frac{n\pi}{3}$

b) solve by Cramer's rule $\frac{1}{x}+\frac{2}{y}-\frac{1}{z}=1;\ \frac{2}{x}+\frac{4}{y}+\frac{1}{z}=5;\ \frac{3}{x}-\frac{2}{y}-\frac{2}{z}=0$

23 a) solve $2\tan^{-1}(\cos x)=\tan^{-1}(2\operatorname{cosec}x)$ (OR)

b) if the roots of $x^3+px^2+qx+r=0$ are in HP prove that $9pqr=27r^2+2q^3$

24 a) $2+I$ and $3-\sqrt{2}$ are roots of the equation $x^6-13x^3+62x^4-126x^3+65x^2+127x-140=0$ (OR)

b) Find the equation of ellipse whose centre(3,-1) one of the foci (6,-1) and passing through (8,-1)

25a) Prove that $2\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{7}=\tan^{-1}\frac{31}{17}$ (OR)

b) Identify the conic and find the centre ,foci, vertices and directrices of $9x^2-4y^2-36x+32y+8=0$

26a) Prove by vector method that the altitudes from the vertices to the opposite sides of a triangle are concurrent. (OR)

b)Graph the tangent and its inverse tangent

27a)Find the parametric and cartesian equations of the plane passing through the non collinear points (3,6,-2), (-1,-2,6) and (6,4,-2) (OR)

b) If line $\vec{a}=\hat{\imath}-\hat{\jmath}$, $\vec{b}=\hat{\imath}-\hat{\jmath}-4\hat{k}$, $\vec{c}=3\hat{\jmath}-\hat{k}$ and $\vec{d}=2\hat{\imath}+5\hat{\jmath}+\hat{k}$ verify that $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})=[\vec{a},\vec{b},\vec{d}]\vec{c}-[\vec{a},\vec{b},\vec{c}]\vec{d}$

www.Padasalai.Net

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com